IO SKY PLATFORM

Milestones 1 to 3 Completed Delivery Record

Verified against the agreed Three Milestone Plan

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What this document is

This is a record of the work across Milestones 1, 2 and 3 that is finished and verified. It lists completed items only. It is not a claim that all three milestones are closed, and the outstanding work is tracked separately in the live milestone checklists.

Everything listed here met the same acceptance bar before it was marked complete: the full TypeScript typecheck passes, automated tests cover the specific behaviour claimed rather than merely the file it lives in, and where the work touches an external system it was demonstrated against a running environment rather than a mock alone.

The AI Scan engine, IVR and AI Receptionist specifications are Phase 4 scope and are deliberately not covered here.

Verification evidence, as at this document's date

TypeScript typecheck	Clean, no errors
Automated test suite	801 tests passing, 0 failures
Test files	73 passing
Production build	Client and server bundles compile
Build artifact secret scan	395 files scanned, no credentials found
Deployment	Live on Railway, container in Amsterdam, database in Ireland

> *One test file cannot run: the live Supabase negative test suite needs the project API key, which was revoked after the credential leak and has not yet been replaced. That is an access item, not a code defect. Every other suite runs green.*

Milestone 1, completed

Milestone 1 covered repository remediation and the migration onto Supabase. The following is complete and verified.

Repository and codebase cleanup

4 items

- Removed 8 checked-in audit output artifacts and the pt_batches directory, 19 files in total
- Deleted 13 confirmed dead code files
- Removed unused packages from the dependency tree
- Deleted empty placeholder folders

Secrets and environment

3 items

- Removed hardcoded fallback secrets and added fail-fast validation, so the application refuses to start on a missing secret rather than running with a default
- Reconciled the environment template against the environment reference documentation
- Removed or flagged 19 documented environment variables that no longer had any effect

Source control and continuous integration

2 items

- Documented the branching and commit convention
- Added the continuous integration workflow

Code hygiene

6 items

- Extracted a shared rate limiter module
- Extracted a shared client IP and user agent extraction module
- Extracted a shared public reference generator
- Removed a duplicated developer gate evaluation in the tRPC layer
- Removed a dead component and resolved a naming collision
- Introduced a shared date formatting helper across 11 call sites

Defects fixed, several security relevant

9 items

- Booking slot release defect
- Reschedule by token performed a 5,000 row scan on every call
- Multi-factor authentication lockout race condition
- Client member redirect drift
- Admin AI Scans key performance indicator reported the wrong figure
- Added alerting when an admin or security audit write fails, so a failed audit record is no longer silent
- Unauthenticated access to stored objects, an insecure direct object reference
- Replaced a non-cryptographic random number generator with a cryptographic one
- Removed personally identifiable information from console logs

Supabase provisioning and database migration

8 items

- Supabase projects provisioned and connection strings recorded
- Schema translated from the MySQL dialect to PostgreSQL
- Foreign keys defined across the schema
- One index per foreign key column, since PostgreSQL does not create these automatically, plus additional indexes on high traffic queries
- Rewrote MySQL specific write path code for PostgreSQL
- Migrated the data
- Row Level Security policies applied

Authentication, access control and tenant isolation

7 items

- Authentication path decision taken and recorded
- Profile design, and migration of login, OAuth, multi-factor authentication and password reset
- Full role based access control ported onto the Supabase session
- Fixed an inconsistency in the solutions admin listing
- Super Admin role decision taken and recorded
- Shared route guard component applied across protected routes
- Negative cross-tenant Row Level Security test suite, which proves one organisation cannot read another's data

Milestone 2, completed

Milestone 2 covered storage migration, removal of the previous vendor platform, and the first substantial platform subsystems. The following workstreams are complete and verified.

Storage migration

- Bucket and path architecture designed and applied
- Application code migrated off the previous vendor's storage proxy onto Supabase Storage
- Official branding assets processed and uploaded to real Supabase Storage, with every resulting URL confirmed publicly reachable before being wired into the application

Removal of the previous vendor platform

- Language model calls moved from the vendor proxy to a direct, provider-neutral integration
- Owner notifications moved onto the platform's own email transport
- Vendor analytics beacon removed
- Vendor build tool plugins and debug collector removed
- Dead code and cosmetic references swept

Email delivery tracking

- Delivery tracking built end to end: schema, logging, provider webhook handling, and admin visibility of delivery status

Core workflow verification and conversion

- Real workflows verified against existing test coverage with no regressions
- The highest severity admin mock screens converted to real data, or honestly disclosed where no backing system existed
- Reports, Projects and Milestones mutations built, plus the AI Scan bridge

Super Admin and platform governance

- Organization Management with role and tenant assignment
- Technical Operator role and the Security Center
- Business Intelligence dashboards

> *These three areas are built, tested and Row Level Security hardened. Other areas inside this workstream, including AI governance configuration and platform configuration, remain outstanding and are tracked in the live checklist.*

Document lifecycle, workflow engine and integration layer

- Document lifecycle with versioning, approval and rejection, and a retention record
- Workflow definition engine
- Integration and webhook registry

> *All three are built, Row Level Security hardened and tested.*

Notification infrastructure

- Central typed notification write service
- Schema extension for notification records
- Mark as read and archive operations
- A working Notification Center on both the Client Portal and the Developer Workspace. The bell on both portals previously showed a real unread count but had no click handler at all, so clicking it did nothing. That is now a working surface.

Milestone 3, completed

Milestone 3 covers deployment, security hardening, testing and production readiness. The following is complete and verified.

Deployment

4 items

- Hosting provider decision taken and recorded
- Production hosting stood up, bound to an explicit port
- Health check endpoint wired into the host
- Confirmed no secret files ship in any build or deploy artifact, enforced by an automated scan in the build itself

Security hardening

8 items

- Security headers applied
- Explicit cross-origin resource sharing policy
- Shared-state rate limiting, so limits hold across multiple running instances rather than per instance
- Session expiry and refresh confirmed as deliberately configured
- Logout confirmed to actually revoke the session rather than only clearing it client side
- Reviewed every use of raw HTML injection in the interface
- Cookie SameSite and Secure attributes reviewed against the real hosting setup
- Input validation coverage audited across the ported procedures

Testing and quality assurance

24 items

This is the largest completed workstream in Milestone 3 and the reason the platform can now be changed with confidence.

- All existing backend tests ported to run against the Supabase backed stack in continuous integration
- React Testing Library introduced, with coverage for the authentication forms, the Notification Center, and every converted admin workflow
- Playwright introduced, with end to end golden paths for login, dashboard, document upload, messaging and booking, per portal
- Milestone 1's negative test patterns formalised into a permanent authentication, access control and tenant isolation suite that runs in continuous integration
- Database suite covering foreign key, Row Level Security and trigger enforcement
- Workflow specific suites for the converted screens including Payments, CRM and Role Management

Defects these suites found and fixed before they reached you

8 items

The testing investment is listed above as process. What it actually produced is below, and this is the part worth reading.

- Double booking was possible in production. The slot hold logic could let two callers take the same slot. Found by a concurrency test, fixed, and now covered by a permanent suite
- The container crash looped on its first deploy. The image built and pushed cleanly, then failed to stay up
- Port scanning fallback would have made the deployment silently unreachable
- All 11 editorial visuals were broken on the live site, across seven pages
- A URI decoding error occurred on every page load
- The users email column had no index at all, so lookups fell back to a full table scan
- A schema and code version mismatch broke every local login
- The send message button was clickable with an empty composer
- An updated-at trigger was missing on three tables

Performance and final verification

2 items

- Confirmed Milestone 1's indexes are genuinely used by the query planner under realistic queries, rather than merely present
- Verified booking slot behaviour under concurrent load

Where this leaves the platform

The platform is deployed and running, the database is migrated and tenant isolated, authentication and access control are rebuilt on Supabase, and the test suite is substantial enough that regressions are caught before they ship. The defect list on the previous page is the clearest evidence of that last point: every one of those issues was found internally rather than by you.

Three items sit outside our control and currently constrain production behaviour:

- The Supabase project API key was revoked after the credential leak and has not been replaced. File storage, document upload and avatars are non-functional in production until it is reissued
- No payment processor credential exists in any environment, which gates AI Scan purchase
- Two Product Owner decisions recorded as open in the Master SRS Appendix D, covering notification template structure and the document governance matrix

A note on the notification specification

The Notification Center Functional Specification and the Notification Event Catalogue version 1.0 have now been received. The catalogue's 84 events across its nine families have been encoded as typed, tested data in the codebase, with the per-family counts asserted against the source document so a silent change fails rather than passing quietly.

The routing engine, delivery pipeline and Notification Center interface that consume that catalogue are not built yet. They are the next substantial piece of Milestone 3 work and they are not represented as complete anywhere in this document.